

AML Assignment 2

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June 14, 2023

1. (1.25 points) Consider $\mathcal{H}$ the following hypothesis class :

$$\mathcal{H} = \{h_a : \mathbb{R} \rightarrow \{0, 1\} \mid a > 0, a \in \mathbb{R}, \text{ where } h_a(x) = 1_{[-a, a]}(x) = \begin{cases} 1, & x \in [-a, a] \\ 0, & x \notin [-a, a] \end{cases}\}$$

- Compute the growth function $\tau_H(m)$ (also known as shatter coefficient) for $m \geq 0$ for hypothesis class $\mathcal{H}$. (1 point)
- Compare your result from the previous point with the general upper bound given by the Sauer lemma. Are they equal or different? (0.25 points)

Solution:

- a. $\mathcal{H}$ can be rewritten as

$$\mathcal{H} = \{h_a : \mathbb{R} \rightarrow \{0, 1\} \mid a > 0, a \in \mathbb{R}, \text{ where } h_a(x) = \begin{cases} 1, & |x| \leq a \\ 0, & |x| > a \end{cases}\}$$

The growth function $\tau_H(m)$ is the maximum number of different functions from a set C of size m to $\{0, 1\}$.

We know that if $\text{VCdim}(\mathcal{H}) = d$ then for any $m \leq d$ we have $\tau_H(m) = 2^m$. Also we need to know the value of d for Sauer lemma. So let's find $\text{VCdim}(\mathcal{H})$.

Obviously $\text{VCdim}(\mathcal{H}) \geq 1$ as we can get both labels 0 and 1. We'll prove $\text{VCdim}(\mathcal{H}) < 2$ by picking $x_1, x_2 \in \mathbb{R}$ such that $|x_1| < |x_2|$. In this case there are 4 labels : 00, 01, 10, 11. For getting 10, $|x_1| > a$ and $|x_2| \leq a \Rightarrow |x_2| \leq a < |x_1| \Rightarrow |x_2| < |x_1| \Rightarrow \Leftarrow$.

So $\text{VCdim}(\mathcal{H}) \geq 1$ and $\text{VCdim}(\mathcal{H}) < 2 \Rightarrow \text{VCdim}(\mathcal{H}) = 1 = d$.

Let $C = \{x_1, x_2, \dots, x_m \mid |x_1| < |x_2| < \dots < |x_m|\}$. The labels we can get respect this format:

$$\overbrace{00\dots 0 \ 11\dots 1}^{\text{label}}, \text{ where } i, j \in \mathbb{N}, i + j = m$$

$i \text{ times } j \text{ times}$

So we count starting from $i = 0, j = m$ until $i = m, j = 0$ which results in $m + 1$ possible labels. To conclude,

$$\tau_H(m) = m + 1.$$

b. According to Sauer lemma,

$$\tau_H(m) \leq \sum_{i=0}^d C_m^i$$

In our case $d = 1$ so $\sum_{i=0}^d C_m^i = C_m^0 + C_m^1 = 1 + m$.

My result from the previous point is equal to the general upper bound given by the Sauer lemma.

2. (1.5 points) Consider the concept class C_a formed by the union of two closed intervals $[a, a+1] \cup [a+2, a+4]$, where $a \in \mathbb{R}$. Give an efficient ERM algorithm for learning the concept class C_a and compute its complexity for each of the following cases:

a. realizable case. (1 point)

b. agnostic case. (0.5 point)

Solution:

My algorithm will run on the training set S . In the realizable case, the algorithm loss on S will be 0, meaning all samples will be labeled correctly. In the agnostic case not all samples in S can be labeled correctly but our algorithm will have the minimum loss.

b) I'll start by presenting the algorithm for the agnostic case. Let's sort S and get $S = \{x_1 < \dots < x_m\}$. This step has complexity $O(m \log m)$. In order to find out the value of a that gives the least mislabeled samples I'll use a sweeping algorithm. I'll try all values for a where the samples belonging to the intervals changes.

Let $0 < \epsilon < \min(x_{i+1} - x_i)$, $i = \overline{1, m-1}$.

I'll start from $a = x_1 - 4 - \epsilon$, which means all points will be labeled as 0 so our initial loss $\min_l = l_i$ will be the points labeled as 1. The complexity of this step is $O(n)$. While increasing a we are interested in the following events (in these order):

1. $x_i \in [a+2, a+4]$, predicted(x_i) = 1
2. $a+1 < x_i < a+2$, predicted(x_i) = 0
3. $x_i \in [a, a+1]$, predicted(x_i) = 1
4. $x_i < a$, predicted(x_i) = 0

Initial event (0) $x_i > a+4$, predicted(x_i) = 0.

The next value for $a = x_i - 4$, x_i will be labeled as 1 and all others as 0, our new loss will be

$$l = \begin{cases} l_i + 1 & \text{if } y_i = 0 \\ l_i - 1 & \text{if } y_i = 1 \end{cases}$$

Let x_{j_0} minimum point in S in event 0. Analogically let $x_{j_1}, x_{j_2}, x_{j_3}$ minimum points in events 1, 2 and 3. The next value for a will be

$$a = \min \begin{cases} x_{j_3} + \epsilon, x_{j_3} \text{ will no longer belong to } [a, a+1] \\ x_{j_2} - 1, x_{j_2} \text{ will start belonging to } [a, a+1] \\ x_{j_1} - 2 + \epsilon, x_{j_1} - 2 \text{ will no longer belong to } [a+2, a+4] \\ x_{j_0} - 4, x_{j_0} \text{ will start belonging to } [a+2, a+4] \end{cases}$$

The algorithm will run until x_m gets to event 4. Now our loss will adjust according to the point whose state changed. If x_j is predicted as 1 and $x_j = 1, l - 1$, else $l + 1$. If x_j is predicted as 0 and $x_j = 0, l - 1$, else $l + 1$. At each step if $l < \min_l$, $\min_l = l$, $res = a$, where res will be our algorithm's result (the value of a with the minimum loss). Of course at each step we need to update $x_{j_0}, x_{j_1}, x_{j_2}, x_{j_3}$.

This part of the algorithm runs in $O(m)$, so the overall complexity will be $O(m \log m)$.

a) Of course we can successfully apply the algorithm from point b) and obtain the value for a with loss 0, but we can use a more efficient algorithm for the realizable case. We go through S once and find $\min_1$ minimum x_{i_1} with $y_{i_1} = 1$. If there is no x_{i_1} with $y_{i_1} = 1$ our algorithm will return $a > \max(x_i)$ (e.g. $\max(x_i) + 1$).

We go through S once again and find $\max_1$ maximum x_{i_2} with $y_{i_2} = 1$ so that $x_{i_2} \leq \min_1 + 1$. In the same run we look for $k_1 = \max(x_{j_1}) < \min_1$. If no k_1 then $k_1 = -\infty$.

We go through S once again and find $\min_2$ minimum x_{i_3} with $y_{i_3} = 1$ so that $x_{i_3} \geq \max_1 + 1$. In the same run we look for $k_2 = \min(x_{j_2}) > \max_1, y_{k_2} = 0$. If we find no $\min_2$ and k_2 then it returns $a \in (k_1, \min_1] \cap [\max_1 - 1, \infty)$. If we find no k_2 we return $(k_1 - 2, \min_1 - 2] \cap [\max_1 - 4, \infty)$. If we find no $\min_1$ then $a \in (k_1, \min_1] \cap [\max_1 - 1, k_2 - 1]$ if $k_2 - k_1 < 2$ else $(k_1 - 2, \min_1 - 2] \cap [\max_1 - 4, k_2 - 4)$.

We go through S once again and find $\max_2$ maximum x_{i_4} with $y_{i_4} = 1$. In the same run we look for $k_3 = \max(x_{j_3}) < \min_2$.

Another run to look for $k_4 = \min(x_{j_4}) > \max_2$, if we don't find any then $k_4 = +\infty$. Our algorithm will return $a \in (k_1, \min_1] \cap [\max_1 - 1, k_2 - 1] \cap (k_3 - 2, \min_2 - 2] \cap [\max_2 - 4, k_4 - 4)$. We go through S 5 times so the time complexity is $O(m)$.

3. (1.25 points) Consider a modified version of the AdaBoost algorithm that runs for exactly three rounds as follows:

- the first two rounds run exactly as in AdaBoost (at round 1 we obtain distribution $\mathbf{D}^{(1)}$, weak classifier h_1 with error ϵ_1 ; at round 2 we obtain distribution $\mathbf{D}^{(2)}$, weak classifier h_2 with error ϵ_2).
- in the third round we compute for each $i = 1, 2, \dots, m$:

$$\mathbf{D}^{(3)}(i) = \begin{cases} \frac{\mathbf{D}^{(1)}(i)}{Z}, & \text{if } h_1(x_i) \neq h_2(x_i) \\ 0, & \text{otherwise} \end{cases}$$

where Z is a normalization factor such that $\mathbf{D}^{(3)}$ is a probability distribution.

- obtain weak classifier h_3 with error ϵ_3 .
- output the final classifier $h_{final}(x) = \text{sign}(h_1(x) + h_2(x) + h_3(x))$.

Assume that at each round $t = 1, 2, 3$ the weak learner returns a weak classifier h_t for which the error ϵ_t satisfies $\epsilon_t \leq \frac{1}{2} - \gamma_t, \gamma_t > 0$.

- What is the probability that the classifier h_1 (selected at round 1) will be selected again at round 2? Justify your answer. (0.25 points)
- Consider $\gamma = \min\{\gamma_1, \gamma_2, \gamma_3\}$. Show that the training error of the final classifier h_{final} is at most $\frac{1}{2} - \frac{3}{2}\gamma + 2\gamma^3$ and show that this is strictly smaller than $\frac{1}{2} - \gamma$. (1 point)

Solution:

a. Let h_1 be the weak classifier selected at round 1 with loss $\epsilon_1 < \frac{1}{2}$. In distribution $\mathbf{D}^{(1)}$ on $\{1, \dots, m\}$, $\mathbf{D}^{(1)}(i) = \frac{1}{m}$.

If example x_i is correctly classified by h_1 then $h_1(x_i) = y_i$ then

$$\mathbf{D}^{(2)}(i) = \frac{1}{m} \times \frac{\sqrt{\frac{\epsilon_1}{1-\epsilon_1}}}{Z_2}.$$

If example x_i is misclassified then $h_1(x_i) \neq y_i$ then

$$\mathbf{D}^{(2)}(i) = \frac{1}{m} \times \frac{\sqrt{\frac{1-\epsilon_1}{\epsilon_1}}}{Z_2}.$$

Let $n \in \mathbb{N}, n \leq m$ be the number of misclassified samples by h_1 . Then

$$\frac{n}{m} \times \frac{\sqrt{\frac{1-\epsilon_1}{\epsilon_1}}}{Z_2} + \frac{m-n}{m} \times \frac{\sqrt{\frac{\epsilon_1}{1-\epsilon_1}}}{Z_2} = 1 \Leftrightarrow$$

$$Z_2 = \epsilon_1 \times \sqrt{\frac{1-\epsilon_1}{\epsilon_1}} + (1-\epsilon_1) \times \sqrt{\frac{\epsilon_1}{1-\epsilon_1}} \Leftrightarrow$$

$$Z_2 = \frac{\epsilon_1(1-\epsilon_1)}{\sqrt{\epsilon_1(1-\epsilon_1)}} + \frac{(1-\epsilon_1)\epsilon_1}{\sqrt{\epsilon_1(1-\epsilon_1)}} \Leftrightarrow$$

$$Z_2 = 2 \frac{\epsilon_1(1-\epsilon_1)}{\sqrt{\epsilon_1(1-\epsilon_1)}} \Leftrightarrow$$

$$Z_2 = 2\sqrt{\epsilon_1(1-\epsilon_1)}$$

Using the value of Z_2 we get:

- For misclassified:

$$\mathbf{D}^{(2)}(i) = \frac{1}{m \times 2\epsilon_1}$$

- For well classified:

$$\mathbf{D}^{(2)}(i) = \frac{1}{m \times 2(1-\epsilon_1)}$$

At second round h_1 will classify the samples exactly the same but now the samples have different importances so the error ϵ_2 of h_1 on $\mathbf{D}^{(2)}$ would be

$$\epsilon_2 = \frac{n}{m} \times \frac{\sqrt{\frac{1-\epsilon_1}{\epsilon_1}}}{Z_2} \Leftrightarrow$$

$$\epsilon_2 = \epsilon_1 \times \frac{\sqrt{\frac{1-\epsilon_1}{\epsilon_1}}}{2\sqrt{\epsilon_1(1-\epsilon_1)}} \Leftrightarrow$$

$$\epsilon_2 = \frac{\sqrt{\epsilon_1(1-\epsilon_1)}}{2\sqrt{\epsilon_1(1-\epsilon_1)}} = \frac{1}{2}$$

As all the other classifiers have loss $< \frac{1}{2}$ the probability of h_1 being chosen the second time is 0.

b. Let's start by proving the last requirement $\frac{1}{2} - \frac{3}{2}\gamma + 2\gamma^3 < \frac{1}{2} - \gamma + \gamma - \frac{1}{2}$.

$$-\frac{1}{2}\gamma + 2\gamma^3 < 0$$

$$\gamma(2\gamma^2 - \frac{1}{2}) < 0 : \gamma > 2, \gamma > 0$$

$$\gamma^2 - \frac{1}{4} < 0 : \gamma < \frac{1}{2}$$

$$\gamma^2 < \frac{1}{4}$$

$$\gamma < \frac{1}{2}, \text{ which is true.}$$

In the previous point we proved that in the first two rounds our algorithm will definitely use two different classifiers h_1, h_2 . Let ϵ_{final} be the error of h_{final} . When does h_{final} make errors? When $h_1(x_i) = h_2(x_i) \neq y_i$ or $(h_1(x_i) \neq h_2(x_i) \text{ and } h_3(x_i) \neq y_i)$. So,

$$Pr_{i \sim D^{(1)}}[h_{\text{final}}(x_i) \neq y_i] = Pr_{i \sim D^{(1)}}[h_1(x_i) = h_2(x_i) \neq y_i] + Pr_{i \sim D^{(1)}}[h_1(x_i) \neq h_2(x_i) \text{ and } h_3(x_i) \neq y_i].$$

Let's look into both probabilities into more detail. Previously we proved that for misclassification:

$$D^{(2)}(i) = \frac{1}{m \times 2\epsilon_1} \Leftrightarrow D^{(1)}(i) = 2\epsilon_1 D^{(2)}(i).$$

So,

$$Pr_{i \sim D^{(1)}}[h_1(x_i) = h_2(x_i) \neq y_i] = 2\epsilon_1 Pr_{i \sim D^{(2)}}[h_1(x_i) = h_2(x_i) \neq y_i].$$

As $D^{(3)}(i) = 0$ if $h_1(x_i) = h_2(x_i)$,

$$Pr_{i \sim D^{(1)}}[h_1(x_i) \neq h_2(x_i) \text{ and } h_3(x_i) \neq y_i] = Z \times Pr_{i \sim D^{(3)}}[h_3(x_i) \neq y_i] = Z \times \epsilon_3.$$

Let's compute Z .

$$Z = Pr_{i \sim D^{(1)}}[h_1(x_i) = y_i \text{ and } h_2(x_i) \neq y_i] + Pr_{i \sim D^{(1)}}[h_1(x_i) \neq y_i \text{ and } h_2(x_i) = y_i].$$

$$Pr_{i \sim D^{(1)}}[h_1(x_i) \neq y_i \text{ and } h_2(x_i) = y_i] = Pr_{i \sim D^{(1)}}[h_1(x_i) \neq y_i] - Pr_{i \sim D^{(1)}}[h_1(x_i) = h_2(x_i) \neq y_i] =$$

$$\epsilon_1 - 2\epsilon_1 Pr_{i \sim D^{(2)}}[h_1(x_i) = h_2(x_i) \neq y_i] = \epsilon_1(1 - 2p),$$

where $p = Pr_{i \sim D^{(2)}}[h_1(x_i) = h_2(x_i) \neq y_i]$.

As for well classified $D^{(2)}(i) = \frac{1}{m \times 2(1-\epsilon_1)} \Leftrightarrow D^{(1)}(i) = 2(1-\epsilon_1)D^{(2)}(i)$,

$$\begin{aligned} Pr_{i \sim D^{(1)}}[h_1(x_i) = y_i \text{ and } h_2(x_i) \neq y_i] &= 2(1-\epsilon_1)(Pr_{i \sim D^{(2)}}[h_2(x_i) \neq y_i] - Pr_{i \sim D^{(2)}}[h_1(x_i) \neq y_i \text{ and } h_2(x_i) \neq y_i]) = \\ &= 2(1-\epsilon_1)(\epsilon_2 - p). \end{aligned}$$

Coming back to Z ,

$$\begin{aligned} Z &= \epsilon_1(1-2p) + 2(1-\epsilon_1)(\epsilon_2 - p) \Rightarrow \\ Pr_{i \sim D^{(1)}}[h_1(x_i) \neq h_2(x_i) \text{ and } h_3(x_i) \neq y_i] &= \epsilon_3(\epsilon_1(1-2p) + 2(1-\epsilon_1)(\epsilon_2 - p)). \end{aligned}$$

So,

$$\begin{aligned} \epsilon_{\text{final}} &= 2\epsilon_1 p + \epsilon_3(\epsilon_1(1-2p) + 2(1-\epsilon_1)(\epsilon_2 - p)) \Leftrightarrow \\ \epsilon_{\text{final}} &= 2\epsilon_1 p - 2\epsilon_1 \epsilon_2 \epsilon_3 + \epsilon_1 \epsilon_2 + 2\epsilon_2 \epsilon_3 - 2\epsilon_3 p \end{aligned}$$

We have to compare ϵ_{final} with $\frac{1}{2} - \frac{3}{2}\gamma + 2\gamma^3$.

$$\epsilon_1, \epsilon_2, \epsilon_3 \leq \epsilon \leq \frac{1}{2} - \gamma, \text{ so}$$

$$\begin{aligned} \epsilon_{\text{final}} &\leq 2\epsilon - 2\epsilon^3 + 3\epsilon^2 - 2\epsilon \leq -2\epsilon^3 + 3\epsilon^2 \leq \epsilon^2(-2\epsilon + 3) \\ &\leq \left(\frac{1}{2} - \gamma\right)^2(-2\left(\frac{1}{2} - \gamma\right) + 3) \leq \left(\frac{1}{4} - \gamma + \gamma^2\right)(2 + 2\gamma) \\ &\leq \frac{1}{2} - 2\gamma + 2\gamma^2 + \frac{1}{2}\gamma - 2\gamma^2 + 2\gamma^3 \leq \frac{1}{2} - \frac{3}{2}\gamma + 2\gamma^3. \end{aligned}$$

4. (1 point) Consider H_{2DNF}^d the class of 2-term disjunctive normal form formulae consisting of hypothesis of the form $h: \{0,1\}^d \rightarrow \{0,1\}$,

$$h(x) = A_1(x) \vee A_2(x)$$

where $A_i(x)$ is a Boolean conjunction of literals H_{conj}^d .

It is known that the class H_{2DNF}^d is not efficient properly learnable but can be learned improperly considering the class H_{2CNF}^d . Give a γ -weak-learner algorithm for learning the class H_{2DNF}^d which is not a strong PAC learning algorithm for H_{2DNF}^d (like the one considering H_{2CNF}^d). Prove that this algorithm is a γ -weak-learner algorithm for H_{2DNF}^d .

Hint: Find an algorithm that returns $h(x) = 0$ or the disjunction of 2 literals.

Solution:

$$h(x) = A_1(x) \vee A_2(x) = \bigwedge_{u \in A_1, v \in A_2} (u \vee v).$$

Now H_{2DNF}^d is a conjunction of $(2n)^2$ variables. Let's take a random $u \vee v$.

u	0	0	1	1
v	0	1	0	1
$u \vee v$	0	1	1	1

By assuming $u \vee v = 1$, our error is at most $\frac{1}{4}$.

An algorithm A is a γ -weak-learner for a class H if there is a function $m_H : (0, 1) \rightarrow \mathbb{N}$ such that:

- for every $\delta > 0$
- for every labeling $f \in H, f : X \rightarrow \{0, 1\}$
- for every distribution D over X

when we run A on $S, |S| \leq m_H(\gamma)$, A returns a hypothesis h such that with probability at least $1 - \delta$, $L_{D,f} \leq \frac{1}{2} - \gamma$.